



Mantavya Maan
Mathematics
Indian Institute of Technology Bombay

23B3308
B.S.
Gender: Male
DOB: 17/08/2005

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2027	8.86

Pursuing a minor degree in **Game Theory and Logic** from Computer Science Department, IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Secured top **0.49** percentile in **JEE Main 2023** examination out of **1.1 million** eligible candidates all over India
- Among the top **0.97** percentile in **JEE Advanced 2023** examination out of **0.18 million** candidates all over India

PROFESSIONAL EXPERIENCES

Data Analytics | Summer Intern | Fino Payment Bank

(June'25–July'25)

- Processed and analyzed **1M+ financial records** using SQL and Python, identifying market trends, customer behavior.
- Developed interactive Tableau dashboards visualizing **20+ KPIs** supporting **executive decision-making** systems.
- Collaborated with **cross-functional teams**, enhancing data integrity and contributing to improved analytical models.

Machine Learning & Data Engineering | Gradpipe

(Nov'25–Dec'25)

- Architected **high-dimensional talent discovery engine** using **vector DBs** and hybrid search for semantic matching.
- Engineered **AST-based code analysis pipeline** evaluating complexity, concurrency, and memory for builder scoring.
- Developed **LLM evaluation framework** ensuring consistent benchmarking accuracy using **judge-based evaluation**.

Mathematical Modeling & AI | Summer Intern | Palpx

(May'26–Jul'26)

- Built **4 predictive scoring models** using statistical modeling, probabilistic modeling, and mathematical optimization.
- Applied **Lambda Calculus** in computational research, mathematical modeling, and AI system development **workflows**.
- Built an intelligent **LLM routing engine** supporting **100+ AI models** using utility optimization and model orchestration.

KEY PROJECTS

Neural Orchestrator | Multi-LLM AI Platform

(May'26–Jul'26)

- Architected an **AI orchestration platform** for **100+ LLMs** using Bayesian inference, Thompson sampling, and planning.
- Built a **self-learning semantic parser** integrating **Llama 3.1**, few-shot memory, SQLite feedback, and task decomposition.
- Engineered a **FastAPI control plane** with OpenRouter synchronization, dynamic model registry, and policy routing.
- Developed **multi-stage execution planning** supporting verifier selection, fallback routing, streaming, and audit trails.

Hybrid RAG Framework | Indian Constitution Assistant

(Jun'26–Jul'26)

- Built a **RAG framework** integrating **Qwen2.5:7B**, **nomic-embed-text**, LangChain, ChromaDB, Pinecone, and Ollama.
- Applied **semantic retrieval**, vector embeddings, prompt engineering, and source grounding for **grounded responses**.
- Engineered **dual-mode architecture** using local inference, cloud deployment, and seamless mode switching capabilities.

Lead Score Model | Decision Intelligence Framework

(May'26–Jun'26)

- Built a mathematical lead qualification framework using **10 scoring parameters** spanning **125+ business variables**.
- Applied **Bayesian inference**, Beta confidence intervals, and nonlinear optimization for **robust lead qualification**.
- Engineered **explainable lead prioritization** using **sigmoid functions**, **exponential decay**, calibration, and optimization.

TECHNICAL SKILLS

Languages	C, C++, Java, Python, NumPy, Pandas, Matplotlib, SciPy, scikit-learn, FastAPI, HTML, CSS, JavaScript
Technologies	Git, GitHub, Jupyter Notebook, Tableau, Power BI, Vector Databases, Hybrid Search, LaTeX, JSON, OOP

EXTRACURRICULAR ACTIVITIES

Sports	- Participated and won silver in Fresher' Sports Championship Powerlifting Competition. (Oct'23) - Participated and won bronze medal in the Overall Institute Level Powerlifting Opens. (Mar'24) - Managed 15+ T20 matches during Aahvan, managing schedules, and smooth execution. (May'25)
Misc	- Participated in Equity Research Competition by conducting fundamental analysis on 20+ firms. - Managed hostel maintenance for 500+ residents, improving efficiency and organizing events. - Completed Summer of Science in DSA and Season of Code in Competitive Programming. - Worked as a Services and Operations Coordinator in Mood Indigo, Asia's Largest Cultural Fest. - Worked as an Infrastructure Coordinator at Techfest, IIT Bombay, Asia's Largest Technical Fest.